

JEE MAIN REVISION MASTER: CAPACITOR

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Capacitance & Dielectric Insertion	15 PYQs	Capacitance
Moderate	Capacitor Networks & Energy Redistribution	4 PYQs	Circuits & Energy

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Capacitance & Dielectric Insertion	$C_0 = \frac{\epsilon_0 A}{d}$, $\text{quad } C = \frac{K\epsilon_0 A}{d}$, $\text{quad } C_{\text{slab}} = \frac{\epsilon_0 A}{d(1 - 1/K)}$, $\text{quad } q_p = q\left(1 - \frac{1}{K}\right)$
Capacitor Networks & Energy Redistribution	$C_{\text{parallel}} = \sum C_i$, $\text{quad } \frac{1}{C_{\text{series}}} = \sum \frac{1}{C_i}$, $\text{quad } V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$, $\text{quad } \Delta U_{\text{loss}} = \frac{C_1 C_2}{2(C_1 + C_2)}(V_1 - V_2)^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Capacitance & Dielectric Insertion (Very High Yield — 15 PYQs)

Core Principle: Parallel plate capacitance, dielectric polarization, and partial dielectric slab insertion.

■ **Caution / JEE Pitfall:** If battery remains **CONNECTED**: $V = \text{const}$, C increases by K $\implies$ Q increases by K . If battery is **DISCONNECTED**: $Q = \text{const}$ $\implies$ V decreases by $1/K$.

■ **Speed Trick:** Energy stored $U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$. When dielectric is inserted with disconnected battery, U decreases because electrostatic attraction does work.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is 'd...' [Ans: C]
- [2024 Shift] Q#3 (MCQ): "A capacitor of capacitance $5\mu F$ is charged by a battery of emf $10 V$. At an instant of time, the potential difference across the capacit..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "A parallel plate capacitor of capacitance C_1 with a dielectric slab in between its plates is connected to a battery. It has a potential..." [Ans: N/A]

3.2 Capacitor Networks & Energy Redistribution (Moderate Yield — 4 PYQs)

Core Principle: Equivalent capacitance reduction, charge sharing between interconnected capacitors, and Joule heating loss.

■ **Caution / JEE Pitfall:** When connecting with opposite polarity, use numerator ($C_1 V_1 - C_2 V_2$) for common potential.

■ **Speed Trick:** Energy loss during charge sharing $\Delta U_{\text{(loss)}}$ is completely independent of wire resistance; it always turns into heat and EM radiation.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the circuit shown in the figure, the potential difference across the $4\mu\text{F}$ capacitor is..." [Ans: N/A]
- [2019 Shift] Q#4 (MCQ): "The equivalent capacitance between A and B is..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "Two capacitors of $3\mu\text{F}$ and $6\mu\text{F}$ are connected in series and a potential difference of 900 V is applied across the combination. They ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the circuit shown in the figure, the potential difference across the $4\mu\text{F}$ capacitor is

- (A) 3 V
- (B) 4 V
- (C) 9 V
- (D) 12 V

Correct Answer: None

Solution Analysis: The $9\mu\text{F}$ and $3\mu\text{F}$ capacitors are connected in parallel. In parallel combination, capacitances are added directly. $C_{\text{(p)}} = 9 + 3 = 12\mu\text{F}$ Now this $12\mu\text{F}$ equivalent capacitor is in series with the $4\mu\text{F}$ capacitor. For two capacitors in series, the equivalent capacitance is: $C_{\text{(eq)}} = (4 \times 12 / 4 + 12) = (48 / 16) = 3\mu\text{F}$ The battery voltage is 12V . So the total charge supplied by the battery is: $Q = C_{\text{(eq)}} \times V = 3\mu\text{F} \times 12\text{V} = 36\mu\text{C}$ In a series combination, each capacitor has the same charge. So the charge

Benchmark Problem 2: KCET 2026 (Q#2)

A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is 'd' and area of each plate is 'A', the energy stored in the capacitor is

- (A) $(1/2)\epsilon_0 E^2$
- (B) $\epsilon_0 E a d$
- (C) $(1/2)\epsilon_0 E^2 A d$ [Correct]
- (D) $\frac{E^2 A d \epsilon_0}{2}$

Correct Answer: C

Solution Analysis: For a parallel plate capacitor, the energy stored is $U = (1/2) C V^2$ For a parallel plate capacitor in air or vacuum, $C = \epsilon_0 A / d$ Also, the electric field between the plates is uniform, so $E = (V / d)$ Hence, $V = E d$ Substituting these in the energy formula, $U = (1/2) (\epsilon_0 A / d) (E d)^2$ $U = (1/2) \epsilon_0 A E^2 d$ So, the energy stored in the capacitor is $(1/2) \epsilon_0 E^2 A d$ Correct option: C.